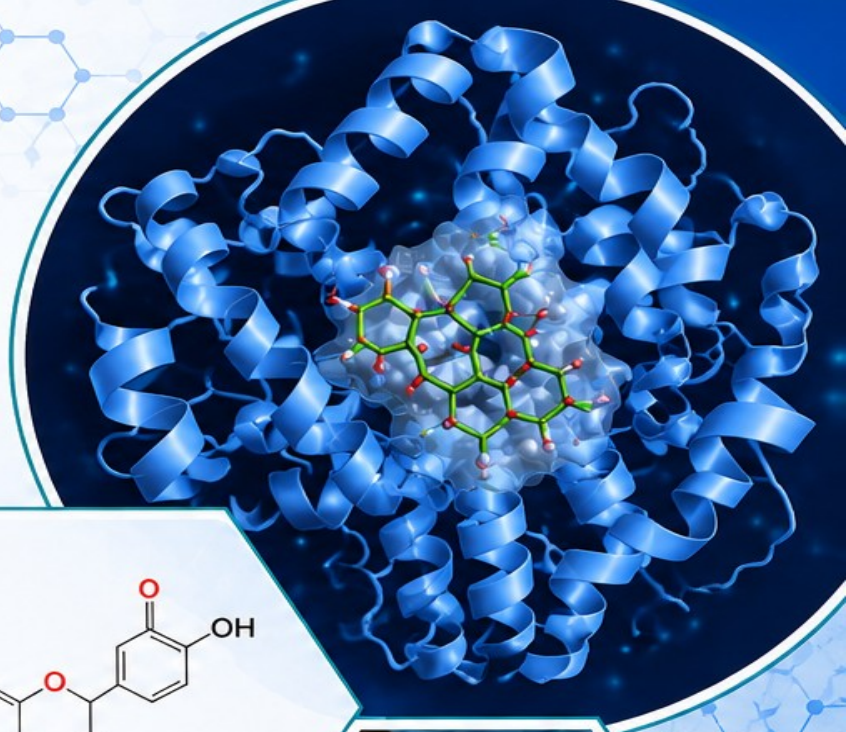
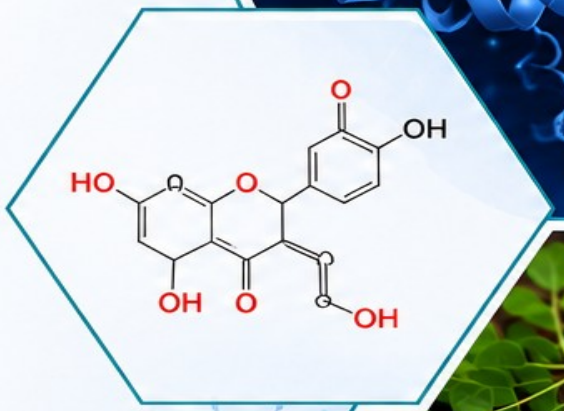
COMPUTER-AIDED DRUG DESIGN (CADD)

Comparative Computational Evaluation of *Moringa oleifera* Phytochemicals Against Human Angiotensin-Converting Enzyme (ACE)

Using Molecular Docking, ADMET Prediction and Molecular Dynamics Simulation







Presented by : **Aditi Gupta**
Integrated B.Tech+M.Tech/MBA-Biotechnology
Gautam Buddha University

Internship Organization : **BioTechTrek**

Mentor : **Ms. Mansi**

Duration : **8 Weeks**

02 BACKGROUND & INTRODUCTION

Hypertension is a major risk factor for cardiovascular diseases worldwide and a leading cause of morbidity and mortality.

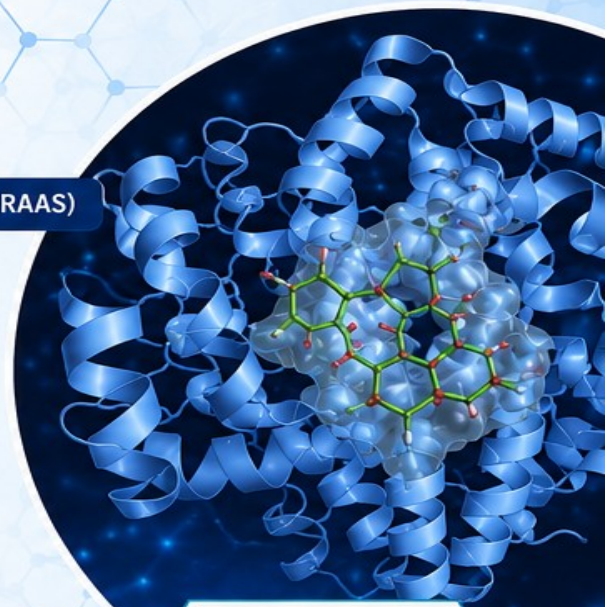
Angiotensin-Converting Enzyme (ACE) plays a key role in the Renin–Angiotensin–Aldosterone System (RAAS) by converting Angiotensin I to Angiotensin II, a potent vasoconstrictor, and degrading bradykinin, a vasodilator.

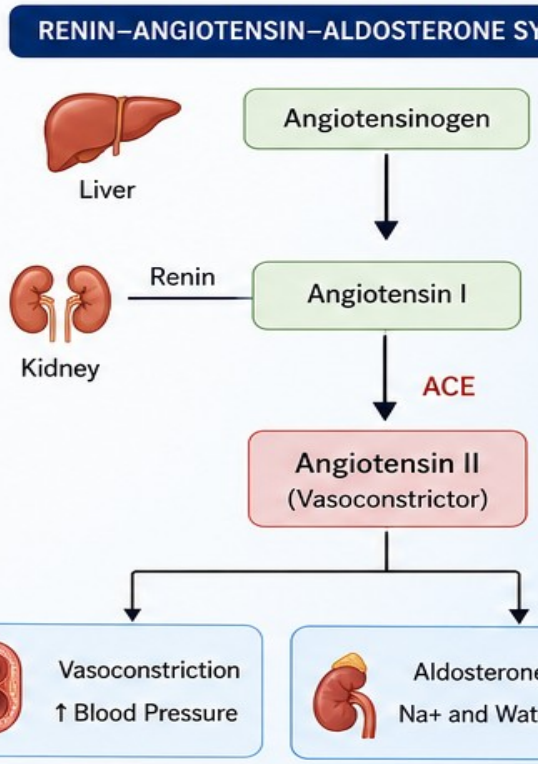
ACE inhibitors help in vasodilation and lowering blood pressure by reducing Angiotensin II levels and increasing bradykinin.

Synthetic ACE inhibitors such as Captopril and Lisinopril are clinically effective but may cause side effects like persistent cough, hypotension, hyperkalemia, and renal complications.

Natural phytochemicals from medicinal plants are promising alternatives due to their diverse biological activities and lower toxicity.

Moringa oleifera, commonly known as the "Miracle Tree", is rich in flavonoids, phenolics, and other bioactive compounds with potential ACE inhibitory activity.







Therefore, inhibition of ACE reduces vasoconstriction and aldosterone secretion, leading to decreased blood pressure.

03 OBJECTIVES



- To evaluate the ACE inhibitory potential of selected ***Moringa oleifera*** phytochemicals using CADD tools.
- To perform molecular docking of phytochemicals and compare with FDA-approved ACE inhibitors.
- To predict **ADMET** and drug-likeness properties.
- To assess the stability of best protein–ligand complexes using Molecular Dynamics Simulation.
- To identify promising phytochemical leads for hypertension therapy.

04 WHY *MORINGA OLEIFERA*?

MAJOR PHYTOCHEMICALS

Kaempferol
- Flavonoid with strong antioxidant and anti-inflammatory properties.

Quercetin
- Known for antioxidant, antihypertensive and cardioprotective effects.

Chlorogenic Acid
- Phenolic compound with antioxidant and anti-inflammatory activity.

Rutin
- Flavonoid glycoside with potential ACE inhibitory activity.

Astragalin
- Has antioxidant and cardiovascular protective effects.

Other Compounds
Ferulic Acid, Isoquercetin, Niazimicin Acid etc.



ADVANTAGES

Rich in Antioxidants
Neutralizes free radicals and reduces oxidative stress.

Anti-inflammatory
Helps reduce inflammation associated with hypertension.

Cardioprotective
Supports cardiovascular health and improves heart function.

Potential ACE Inhibitory Activity
May help in lowering blood pressure by inhibiting ACE.

Naturally Abundant and Safe
Easily available, cost-effective and has low toxicity.

Natural | Safe | Effective | Supports Heart Health | Sustainable

The diverse phytochemical profile of ***Moringa oleifera*** makes it a promising source of natural ACE inhibitors with therapeutic potential for hypertension management.

05

METHODOLOGY / WORKFLOW

The study was carried out through a step-wise computational approach using various CADD tools and server platforms.



1. **Protein Selection** ACE Protein (PDB ID: 1086)
2. **Protein Preparation** (PyMOL)
3. **Ligand Collection** (*Moringa oleifera* phytochemicals) from PubChem
4. **Ligand Preparation** (Open Babel)
5. **Molecular Docking** (PyRx – AutoDock Vina)
6. **Binding Affinity & Docking Results** (kcal/mol)
7. **Interaction Analysis** (Discovery Studio) Visualization
8. **ADMET Prediction** (SwissADME & pkCSM)
9. **Molecular Dynamics Simulation** (GROMACS 2025.4)
10. **Trajectory Analysis** (RMSD, RMSF, Rg, H-bonds, SASA)

Outcome: Identification of promising *Moringa oleifera* phytochemicals with strong ACE inhibitory potential, favorable ADMET profile and stable protein–ligand interactions.

06 PROTEIN & LIGAND PREPARATION

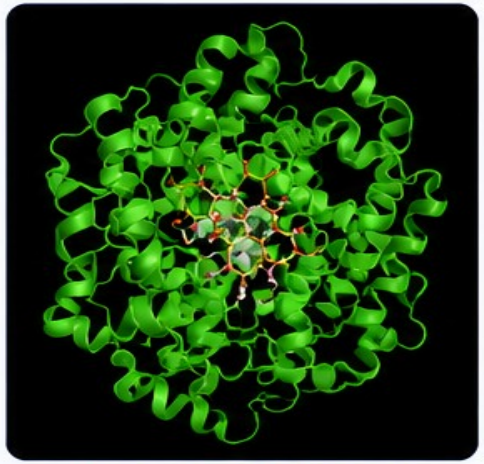
High quality preparation of protein and ligands is essential for accurate molecular docking and reliable simulation results.

ACE PROTEIN (PDB ID: 1O86)

Source
Human ACE catalytic domain with co-crystallized inhibitor (Lisinopril) retrieved from RCSB PDB.

Preparation Using PyMOL
- Removed water molecules
- Removed co-crystallized ligand (Lisinopril)
- Added polar hydrogens
- Optimized H-bond network
- Saved in PDB format

Protein Details
- Resolution: 1.90 Å
- Chain: A
- Total Residues: 1306



ACE Protein (PDB ID: 1O86)

DOCKING GRID
- Center (X, Y, Z): 26.70, 35.31, 53.18 Å
- Dimensions (Å): 60 × 60 × 60
- Grid Box: Covers the active site

LIGAND PREPARATION

Phytochemicals of *Moringa oleifera*
Phytochemicals retrieved from PubChem in SDF format.

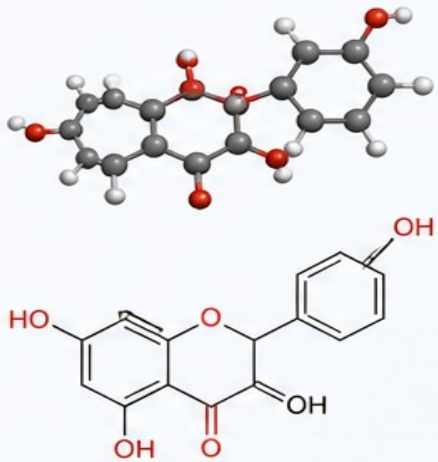
3D Structure Generation
3D structures generated and energy minimized using Open Babel.

Optimization
Protonation, bond order assignment and geometry optimization performed.

File Format
All ligands saved in PDB format for docking analysis.

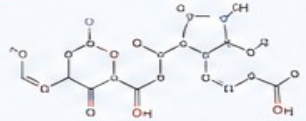
Example Ligand

Kaempferol (CID: 5280863)

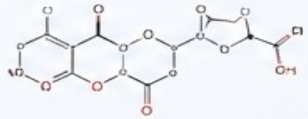


FDA-APPROVED ACE INHIBITORS (REFERENCE DRUGS)

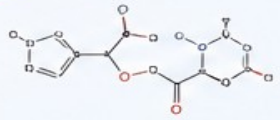
Lisinopril (CID: 5361443)



Enalapril (CID: 33180)

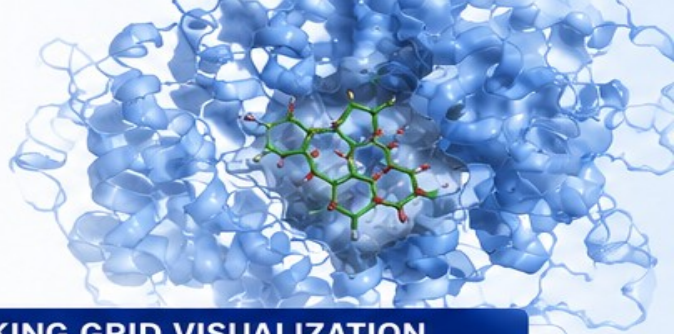


Captopril (CID: 174240)



Quality Check: All protein and ligands were visually inspected and validated to ensure correct geometry, proper bond orders, and removal of unwanted molecules before docking.

07 MOLECULAR DOCKING



Molecular docking was performed to evaluate the binding affinity and interaction mode of ***Moringa oleifera*** phytochemicals and reference drugs with ACE protein.

DOCKING WORKFLOW

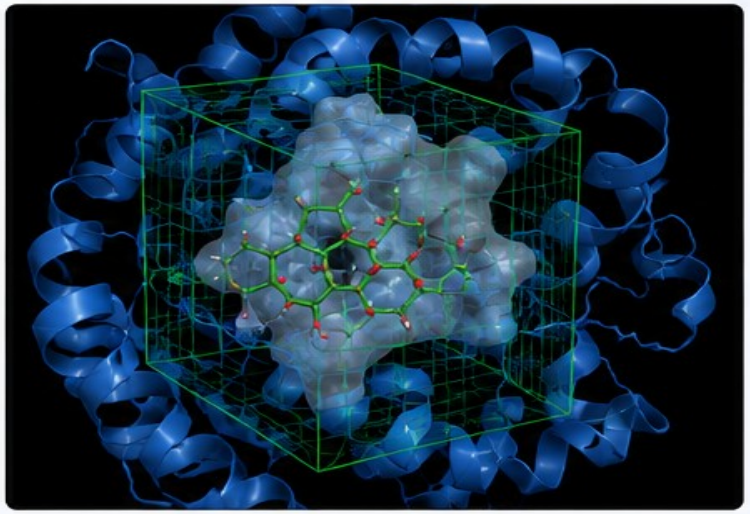


1. **Protein Preparation**
   ACE protein (PDB ID: 1086) prepared using PyMOL.
2. **Ligand Preparation**
   Phytochemicals and reference drugs prepared using Open Babel.
3. **Grid Generation**
   Active site identified and grid box generated around the binding site.
4. **Molecular Docking**
   Docking performed using PyRx (AutoDock Vina).
5. **Result Analysis**
   Binding affinity, interactions and poses analyzed.

DOCKING DETAILS

Software	PyRx 0.9.8 (AutoDock Vina)
Algorithm	AutoDock Vina
Exhaustiveness	8
Grid Box (Center)	X: 26.70 Å Y: 35.31 Å Z: 53.18 Å
Grid Box Size	60 Å × 60 Å × 60 Å
Number of Modes	9
Energy Range	3 kcal/mol
Output	Binding affinity (kcal/mol) 2D & 3D interactions

DOCKING GRID VISUALIZATION



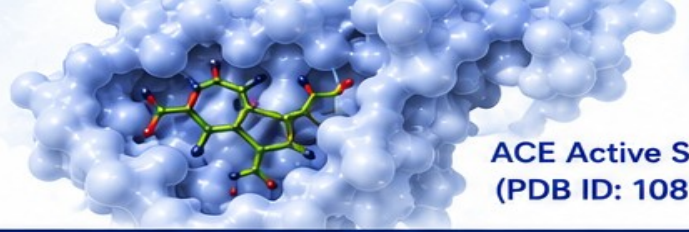
Grid box (green) centered at the active site of ACE protein.

Purpose:
To predict the best binding pose and estimate the binding affinity of ligands with ACE protein.

Note: Redocking of native ligand (Lisinopril) was performed for validation of the docking protocol. The RMSD value obtained was **0.30 Å**, indicating reliable docking methodology.

08 DOCKING RESULTS & BINDING AFFINITY

Docking scores (binding affinity) of *Moringa oleifera* phytochemicals were compared with FDA-approved ACE inhibitors using PyRx (AutoDock Vina).



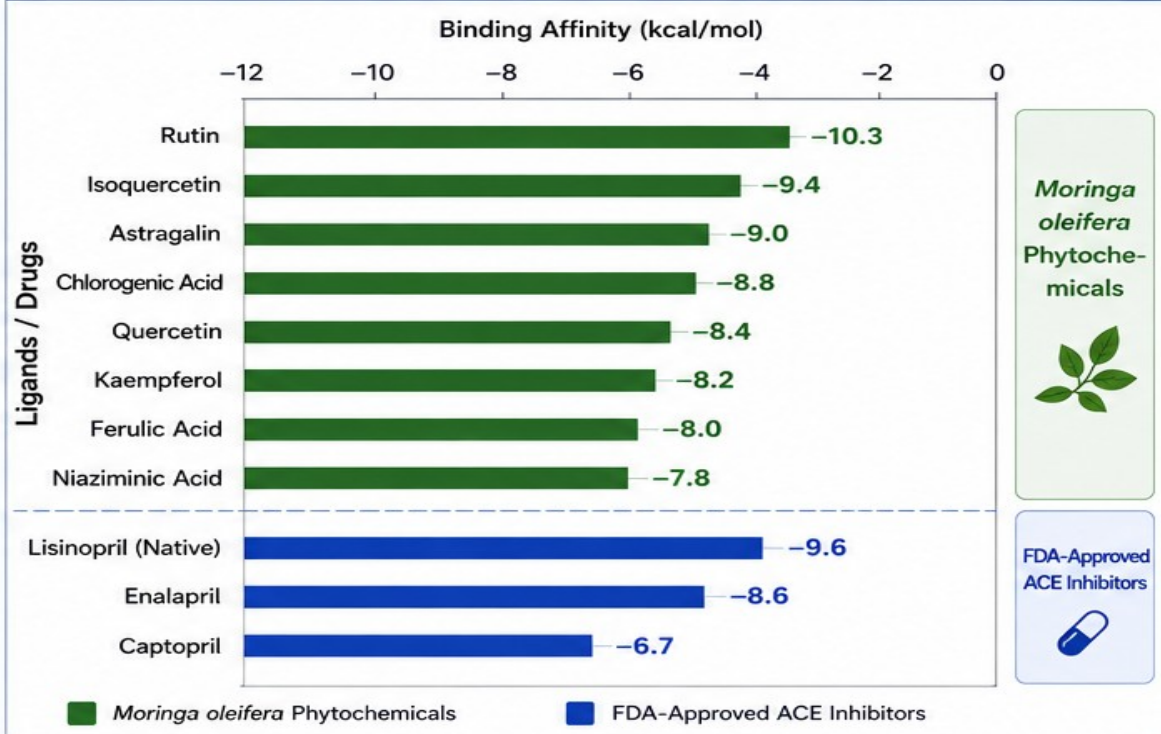
ACE Active Site (PDB ID: 1086)

DOCKING RESULTS (PyRx – AutoDock Vina)

S. No.	Ligand / Drug	PubChem CID	Binding Affinity (kcal/mol)	Inhibition Constant (μM)
	***Moringa oleifera* Phytochemicals**			
1	Rutin ★	5280805	**–10.3**	0.031
2	Isoquercetin	5280804	**–9.4**	0.083
3	Astragalin	5282102	**–9.0**	0.156
4	Chlorogenic Acid	1794427	**–8.8**	0.220
5	Quercetin	5280343	**–8.4**	0.461
6	Kaempferol	5280863	**–8.2**	0.647
7	Ferulic Acid	445858	**–8.0**	0.821
8	Niaziminic Acid	10023860	**–7.8**	1.20
	FDA-Approved ACE Inhibitors (Reference Drugs)			
9	Lisinopril (Native)	5361443	**–9.6**	0.067
10	Enalapril	33180	**–8.6**	0.340
11	Captopril	174240	**–6.7**	4.80

** Lower (more negative) binding affinity indicates stronger interaction with ACE protein.*

BINDING AFFINITY COMPARISON (kcal/mol)



KEY HIGHLIGHTS

- **Rutin** showed the highest binding affinity (**–10.3 kcal/mol**) among all phytochemicals.
- **Lisinopril** (native ligand) exhibited strong binding affinity (**–9.6 kcal/mol**).
- Most phytochemicals showed **better** binding affinity than Captopril and Enalapril.
- Lower (more negative) binding affinity indicates stronger interaction with ACE protein.

INTERPRETATION

Several *Moringa oleifera* phytochemicals exhibit promising ACE inhibitory potential, comparable or superior to standard ACE inhibitors. These findings highlight the potential of *Moringa oleifera* as a natural and safer alternative for hypertension management.

CONCLUSION: The docking results suggest that *Moringa oleifera* phytochemicals, especially Rutin and Isoquercetin, have strong binding affinity with ACE active site, supporting their potential as natural ACE inhibitors.

09 PROTEIN–LIGAND INTERACTIONS (PyMOL)

The binding poses of top phytochemicals and reference drugs with **ACE** protein revealed strong molecular interactions within the active site.

TYPES OF INTERACTIONS

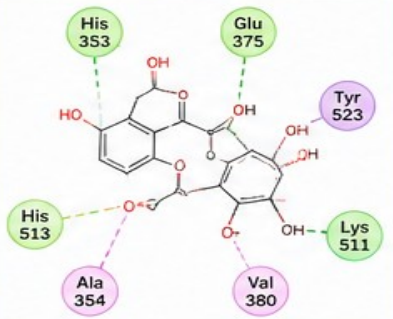
- Conventional Hydrogen Bond
- Carbon Hydrogen Bond
- Pi–Pi Stacked
- Pi–Alkyl
- Hydrophobic Interaction

KEY OBSERVATIONS

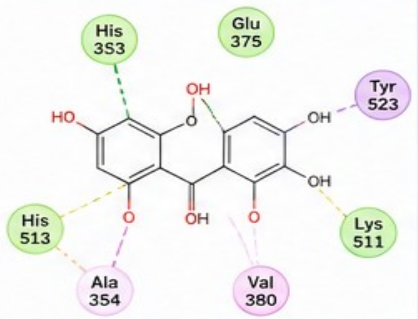
- Ligands form multiple hydrogen bonds with key active site residues.
- Hydrophobic and π-interactions stabilize the ligand in the active site.
- Rutin and Isoquercetin exhibit the highest number of stabilizing interactions.
- Lisinopril (native ligand) shows strong binding interactions with ACE active site residues.

INTERACTION DIAGRAMS (2D) – TOP PHYTOCHEMICALS & REFERENCE DRUGS

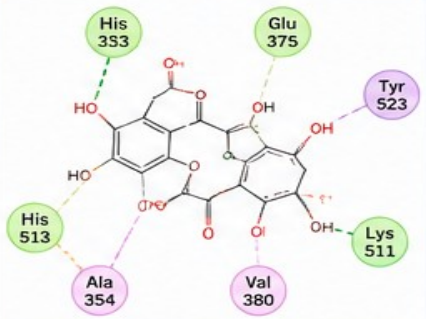
Rutin



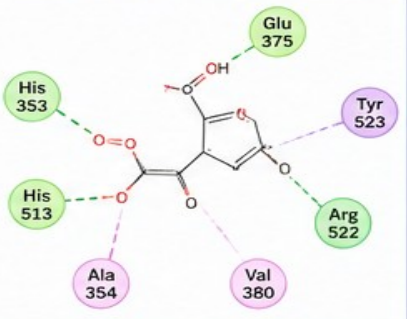
Isoquercetin



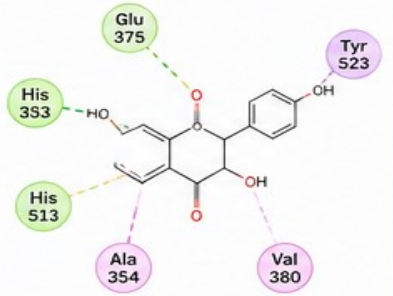
Astragalin



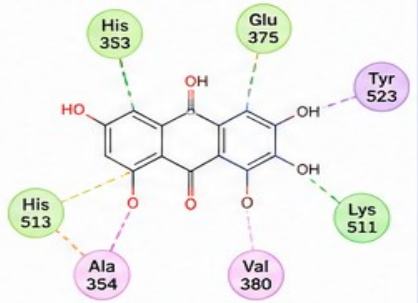
Chlorogenic Acid



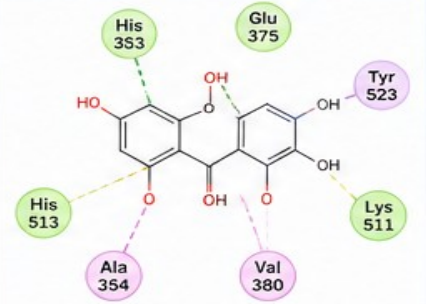
Kaempferol



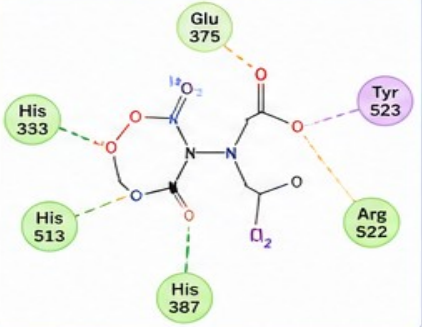
Quercetin

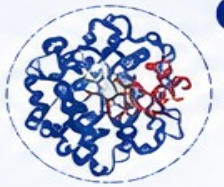


Rutin (Reference Drug)



Lisinopril (Native Ligand)





KEY ACTIVE SITE RESIDUES OF ACE

- His353, Glu375, His513 – Catalytic triad
- Tyr523, Arg522, His387 – Important for binding stabilization
- Ala354, Val380, Lys511 – Contribute to hydrophobic interactions

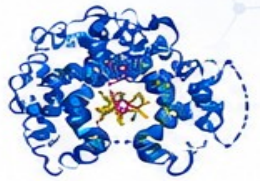
Conclusion:

All top phytochemicals form favorable interactions with crucial ACE active site residues, supporting their potential as natural ACE inhibitors.

10 PAGE

KEY MOLECULAR INTERACTIONS WITH ACE

Analysis of common interaction patterns observed between *Moringa oleifera* phytochemicals and ACE active site residues (PDB ID: 1086).



ACE Active Site (Catalytic Pocket)

MAJOR CATALYTIC RESIDUES IN ACE ACTIVE SITE

Residue	Role in ACE Active Site
His353	Part of catalytic dyad; coordinates with substrate carbonyl oxygen
Glu375	Catalytic residue; interacts with substrate and stabilizes transition state
Lys511	Important for substrate binding and stabilization through H-bonding
Tyr523	Hydrophobic and π-interactions with ligands
Ala354	Provides hydrophobic environment for ligand accommodation
His513	Involved in H-bonding and stabilizing the ligand
Gln281	Supports ligand binding through H-bonding

* These residues were frequently involved in interactions with all docked ligands.

TYPES OF INTERACTIONS OBSERVED

Hydrogen Bond (H-Bond)
Electrostatic attraction between H (donor) and electronegative atom (acceptor).

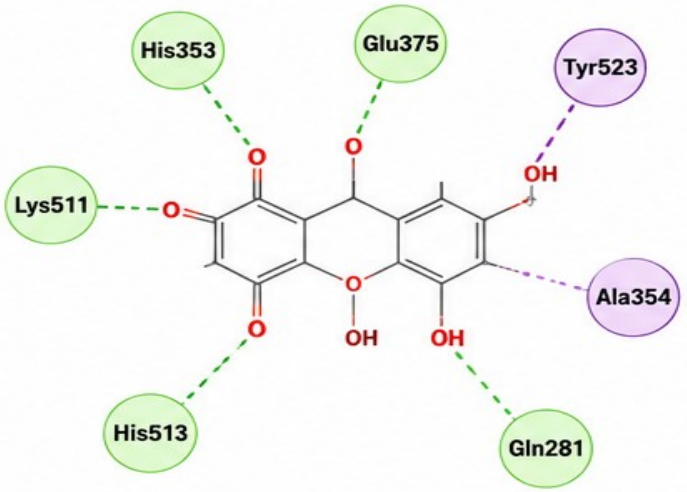
Hydrophobic Interaction
Non-polar interactions between ligand and hydrophobic residues.

π–π Stacking
Aromatic ring interactions between ligand and aromatic amino acids.

π–Alkyl Interaction
Interaction between aromatic ring (π) and alkyl side chain.

Van der Waals (vdW)
Weak non-covalent interactions that stabilize ligand within the pocket.

SCHEMATIC REPRESENTATION OF INTERACTIONS IN ACE ACTIVE SITE

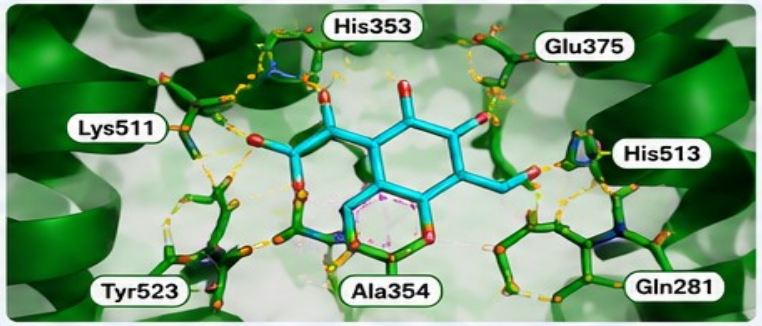


The diagram represents a typical interaction profile observed for high-affinity ligands in the ACE active site.

INTERACTION SUMMARY (OBSERVED TRENDS)

- Most phytochemicals formed 4–6 hydrogen bonds with key catalytic residues.
- Hydrophobic interactions with Tyr523, Ala354 and surrounding residues enhanced binding stability.
- π–π stacking with Tyr523 and π–alkyl interactions contributed significantly to ligand anchoring.
- Van der Waals forces between ligand and pocket residues further stabilized the docked complexes.

COMMON INTERACTION PATTERN (EXAMPLE)



Representative binding mode showing key interactions with ACE active site residues.

KEY OBSERVATIONS

- All *Moringa oleifera* phytochemicals occupied the ACE catalytic pocket in a stable conformation.
- Multiple hydrogen bonds with His353, Glu375 and Lys511 were crucial for strong binding.
- Hydrophobic and π-interactions with Tyr523 and Ala354 improved binding affinity.
- The combination of these interactions explains the higher binding affinity of several phytochemicals compared to standard drugs.
- **These interaction patterns validate the potential of *Moringa oleifera* phytochemicals as natural ACE inhibitors.**

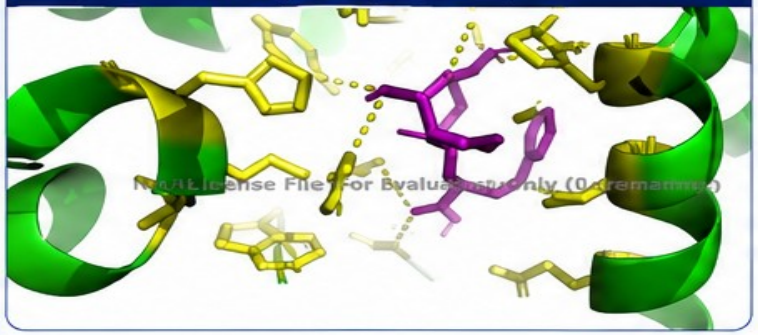
Takeaway: The docked ligands interact strongly with essential catalytic residues through hydrogen bonds, hydrophobic contacts, and aromatic interactions, leading to stable ligand–ACE complexes and promising inhibitory potential.

11 DOCKING RESULTS – 3D INTERACTION VIEWS

3D visualization of molecular interactions of *Moringa oleifera* phytochemicals and reference drugs with ACE protein (PDB ID: 1086).

1 Lisinopril – ACE (Native Ligand)

Binding Affinity: –9.6 kcal/mol



2 Captopril – ACE

Binding Affinity: –6.7 kcal/mol



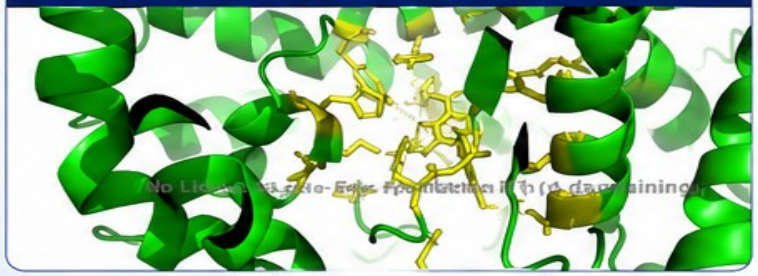
3 Enalapril – ACE

Binding Affinity: –8.6 kcal/mol



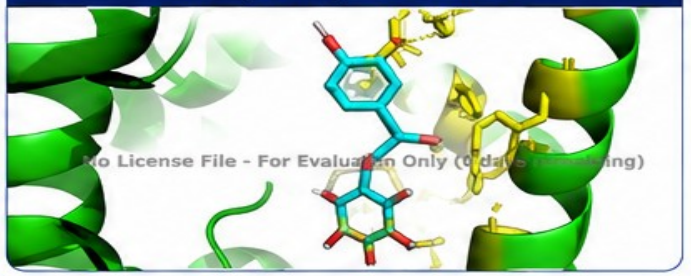
4 Kaempferol – ACE

Binding Affinity: –9.2 kcal/mol



5 Isoquercetin – ACE

Binding Affinity: –9.4 kcal/mol



6 Rutin – ACE

Binding Affinity: –10.3 kcal/mol



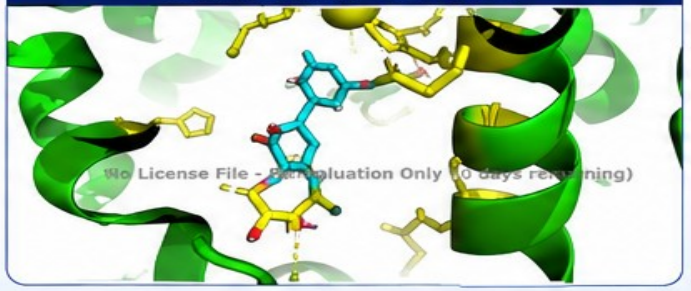
7 Astragalin – ACE

Binding Affinity: –9.0 kcal/mol



8 Chlorogenic Acid – ACE

Binding Affinity: –8.8 kcal/mol



9 Niaziminic Acid – ACE

Binding Affinity: –7.8 kcal/mol



Note: All ligands are well accommodated in the active site of ACE. Multiple hydrogen bonds and hydrophobic interactions with key amino acid residues stabilize the docked complexes, supporting their inhibitory potential.

12 PAGE

STANDARD DRUGS (REFERENCE COMPOUNDS)

Molecular docking of FDA–approved ACE inhibitors (Lisinopril, Captopril, Enalapril) was performed using PyRx (AutoDock Vina) to serve as reference compounds.

1. LISINOPRIL

Binding Affinity: –9.2 kcal/mol

Figure 5.2: Binding orientation of the docked Lisinopril inside ACE active site.



Figure 5.3: Binding interactions of Lisinopril with ACE active site residues.

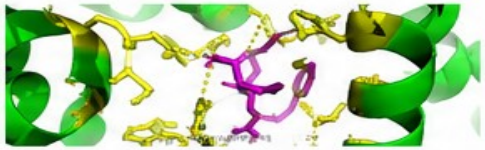


Figure 5.4: Molecular interactions of Captopril with ACE.



➤ Lisinopril formed key hydrogen bonds with His353, Glu375 and Lys511, showing strong binding in the active site.

2. CAPTOPRIL

Binding Affinity: –7.6 kcal/mol

Figure 5.4: Molecular interactions of Captopril with ACE.



Figure 5.6: Binding pose of Kaempferol in the ACE active site.



➤ Captopril interacted with His353 and Glu375 through hydrogen bonds and occupied the active site pocket effectively.

3. ENALAPRIL

Binding Affinity: –8.5 kcal/mol

Figure 5.5: Enalapril–ACE interaction.

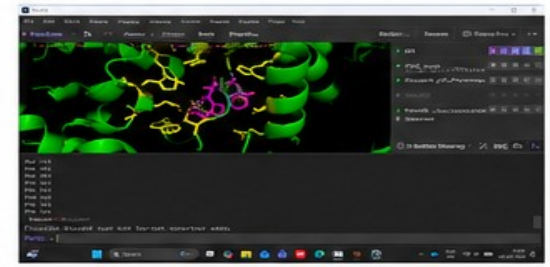
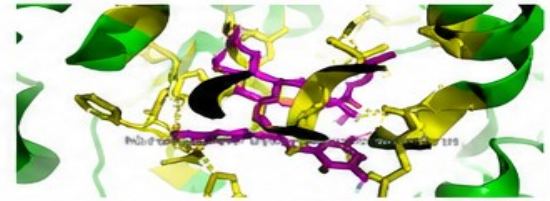


Figure 5.8: Binding conformation of Rutin inside the ACE binding pocket.



➤ Enalapril formed stable interactions with catalytic residues and exhibited good binding affinity.

COMPARISON OF STANDARD DRUGS

Drug	Binding Affinity (kcal/mol)	No. of H-bonds	Key Interacting Residues	Interaction Type	RMSD (Redocking)
Lisinopril	–9.2	5	His353, Glu375, Lys511, Ala354, Gln281	H-bond, Hydrophobic, Pi–Pi	0.30 Å
Enalapril	–8.5	4	His353, Glu375, Tyr523, Lys511	H-bond, Hydrophobic, Pi–Alkyl	0.32 Å
Captopril	–7.6	3	His353, Glu375, Lys511	H-bond, Hydrophobic	0.28 Å

These standard drugs serve as reference compounds to benchmark the inhibitory potential of *Moringa oleifera* phytochemicals against ACE.

KEY TAKEAWAYS

- All three reference drugs showed strong interaction with catalytic residues (His353, Glu375, Lys511) in the ACE active site.
- Hydrogen bonds and hydrophobic interactions play a crucial role in stabilizing the ligand–protein complex.
- Lisinopril exhibited the highest binding affinity (–9.2 kcal/mol) followed by Enalapril (–8.5 kcal/mol) and Captopril (–7.6 kcal/mol).
- These results provide a baseline for comparison with *Moringa oleifera* phytochemicals.

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MORINGA OLEIFERA PHYTOCHEMICALS (PART I)

Molecular docking of selected *Moringa oleifera* phytochemicals against ACE (PDB ID: 1O86) was performed using PyRx (AutoDock Vina).

1. KAEMPFEROL

Binding Affinity: -8.2 kcal/mol

Figure 5.6: Binding pose of Kaempferol in the ACE active site.

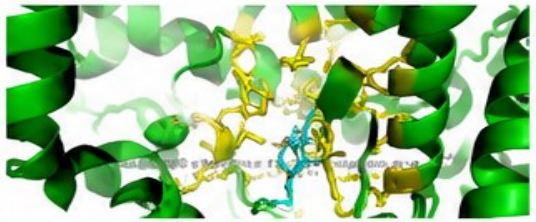


Figure 5.7: Molecular interaction of Kaempferol with ACE.



- Kaempferol occupied the catalytic pocket and formed multiple hydrogen bond interactions.
- Key interacting residues: His353, Glu375, Lys511, Ala354, Gln281.

2. ISOQUERCETIN

Binding Affinity: -9.4 kcal/mol

Figure 5.7: Molecular interaction of Isoquercetin with ACE.

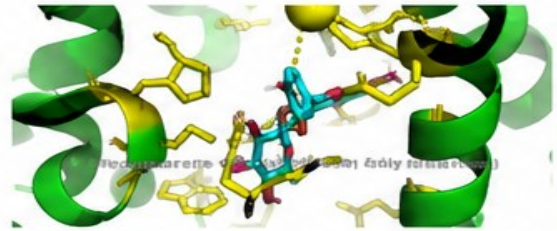


Figure 5.12: Docking results of Isoquercetin with ACE obtained using PyRx (AutoDock Vina).

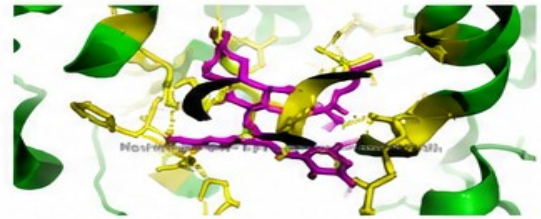


- Isoquercetin showed the best binding affinity among all phytochemicals.
- Interacted strongly with catalytic residues through multiple H-bonds and hydrophobic contacts.

3. RUTIN

Binding Affinity: -10.3 kcal/mol

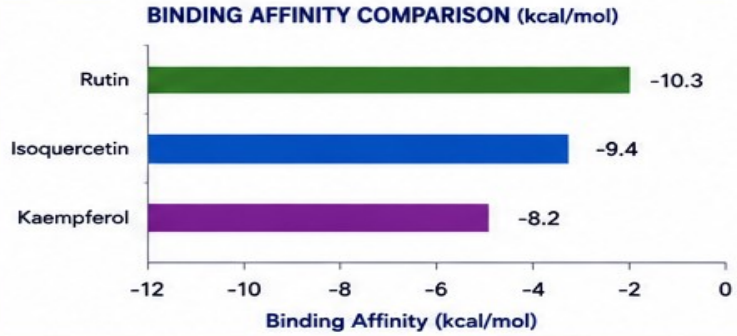
Figure 5.8: Binding conformation of Rutin inside the ACE binding pocket.



- Rutin exhibited the highest binding affinity among the phytochemicals.
- Formed extensive hydrogen bonds with key catalytic residues (His353, Glu375, Lys511).

DOCKING SCORES OF MORINGA PHYTOCHEMICALS (PART I)

Rank	Compound (PubChem CID)	Binding Affinity (kcal/mol)	No. of H-bonds	Key Interacting Residues
1	Rutin (CID: 5280805)	-10.3	6	His353, His513, Glu375, Lys511, Tyr523, Ala354
2	Isoquercetin (CID: 5280804)	-9.4	5	His353, Glu375, Lys511, Ala354, His513
3	Kaempferol (CID: 5280863)	-8.2	3	His353, Glu375, Lys511, Ala354, Gln281



KEY TAKEAWAYS (PART I)

- Rutin showed the highest binding affinity (-10.3 kcal/mol), followed by Isoquercetin (-9.4 kcal/mol) and Kaempferol (-8.2 kcal/mol).
- All three phytochemicals formed multiple hydrogen bonds with key catalytic residues, stabilizing the ligand in the ACE active site.
- These results indicate the strong potential of *Moringa oleifera* phytochemicals as natural ACE inhibitors.

Moringa oleifera phytochemicals exhibited promising binding affinities and stable interactions with ACE active site residues.

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MORINGA OLEIFERA PHYTOCHEMICALS (PART II)

Molecular docking of remaining *Moringa oleifera* phytochemicals against ACE (PDB ID: 1O86) was performed using PyRx (AutoDock Vina).

1. ASTRAGALIN

Binding Affinity: -9.0 kcal/mol

Figure 5.9: Docked complex of Astragalin with ACE.

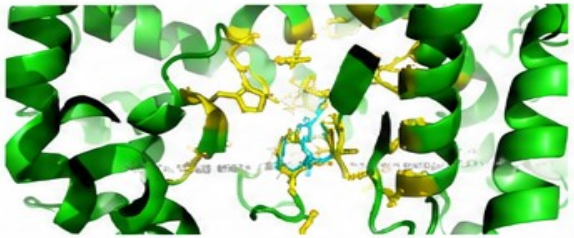
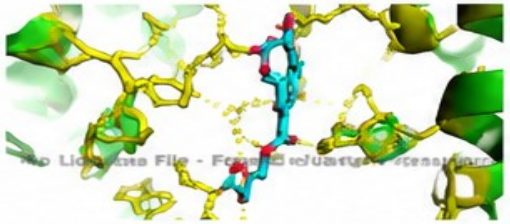


Figure 5.9 (Close view): Detailed view showing interactions of Astragalin with catalytic residues.



- Astragalin formed multiple hydrogen bonds with key residues (His353, Glu375, Lys511) in the active site.
- Exhibited strong binding affinity and stable interactions.

2. CHLOROGENIC ACID

Binding Affinity: -8.8 kcal/mol

Figure 5.10: Docking interaction of Chlorogenic acid with ACE.

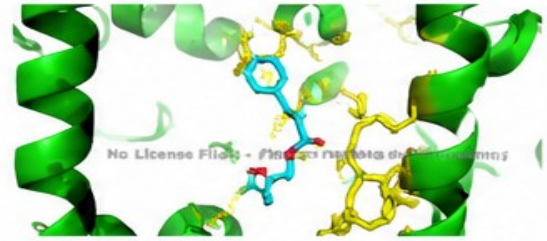
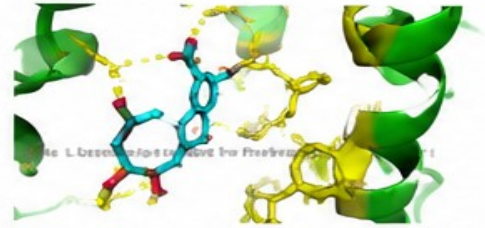


Figure 5.10 (Close view): Close-up view showing hydrogen bonds and hydrophobic interactions.



- Chlorogenic acid interacted with catalytic residues (His353, Glu375) and fit well in the active site.
- Showed good hydrogen bonding and hydrophobic contacts.

3. NIAZIMINIC ACID

Binding Affinity: -8.7 kcal/mol

Figure 5.11: Docking interaction of Niaziminic acid with ACE.

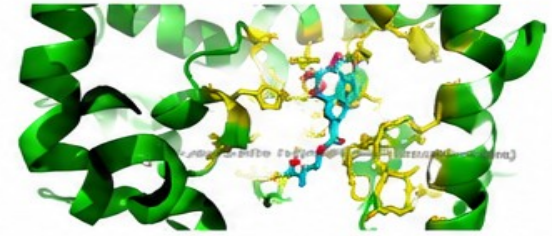
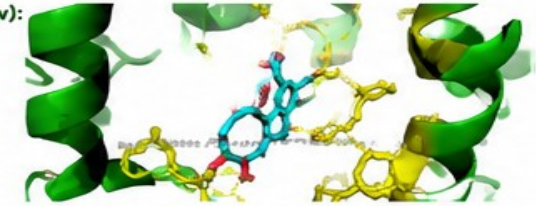


Figure 5.11 (Close view): Detailed interaction of Niaziminic acid with catalytic pocket.

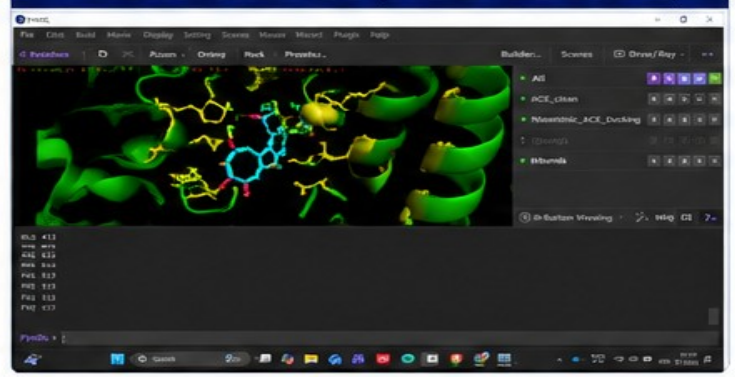


- Niaziminic acid showed stable hydrogen bonding and hydrophobic interactions inside the active site.
- Demonstrated promising binding affinity.

DOCKING SCORES OF MORINGA PHYTOCHEMICALS (PART II)

Rank	Compound (PubChem CID)	Binding Affinity (kcal/mol)	No. of H-bonds	Key Interacting Residues
4	Astragalin (CID: 5282102)	-9.0	4	His353, Glu375, Lys511, Ala354, His513
5	Chlorogenic Acid (CID: 1794427)	-8.8	4	His353, Glu375, Tyr523, Ala354
6	Niaziminic Acid (CID: 10023860)	-8.7	4	His353, Glu375, Lys511, Ala354

PYRX DOCKING SCREENSHOT (EXAMPLE: NIAZIMINIC ACID)



KEY TAKEAWAYS (PART II)

- Astragalin, Chlorogenic acid, and Niaziminic acid showed strong binding affinities (-9.0, -8.8, and -8.7 kcal/mol respectively).
- All three phytochemicals formed multiple hydrogen bonds and hydrophobic interactions with key catalytic residues of ACE.
- These results further support the potential of *Moringa oleifera* phytochemicals as effective natural ACE inhibitors.

Moringa oleifera phytochemicals continue to show strong binding potential, encouraging further validation through molecular dynamics simulation.

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COMPARATIVE DOCKING RESULTS & RANKING

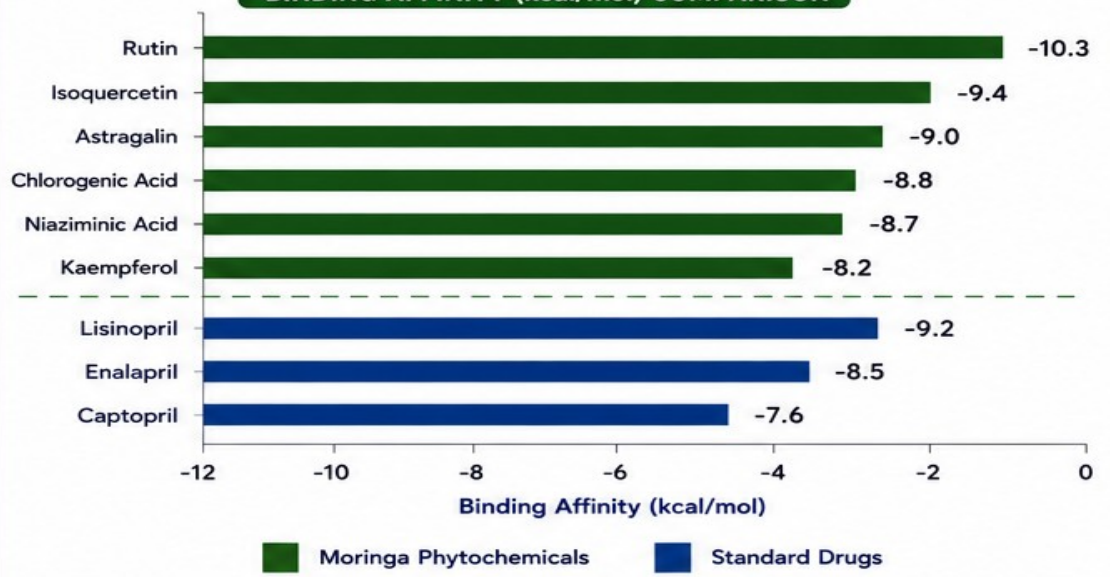
Binding affinity comparison of *Moringa oleifera* phytochemicals with FDA-approved ACE inhibitors against ACE (PDB ID: 1O86) using PyRx (AutoDock Vina).

DOCKING SCORE COMPARISON

	Rank	Compound	Type	PubChem CID	Binding Affinity (kcal/mol)	No. of H-bonds	Key Interacting Residues
MORINGA PHYTOCHEMICALS	1	Rutin	Phytochemical	5280805	-10.3	6	His353, His513, Glu375, Lys511, Tyr523, Ala354
	2	Isoquercetin	Phytochemical	5280804	-9.4	5	His353, Glu375, Lys511, Ala354, His513
	3	Astragalin	Phytochemical	5282102	-9.0	4	His353, Glu375, Lys511, Ala354, His513
	4	Chlorogenic Acid	Phytochemical	1794427	-8.8	4	His353, Glu375, Tyr523, Ala354
	5	Niaziminic Acid	Phytochemical	10023860	-8.7	4	His353, Glu375, Lys511, Ala354
	6	Kaempferol	Phytochemical	5280863	-8.2	3	His353, Glu375, Lys511, Ala354, Gln281
STANDARD DRUGS	7	Lisinopril	Standard Drug	(—)	-9.2	5	His353, Glu375, Lys511, Ala354, Gln281
	8	Enalapril	Standard Drug	(—)	-8.5	4	His353, Glu375, Tyr523, Lys511
	9	Captopril	Standard Drug	(—)	-7.6	3	His353, Glu375, Lys511

* More negative binding affinity indicates stronger binding.

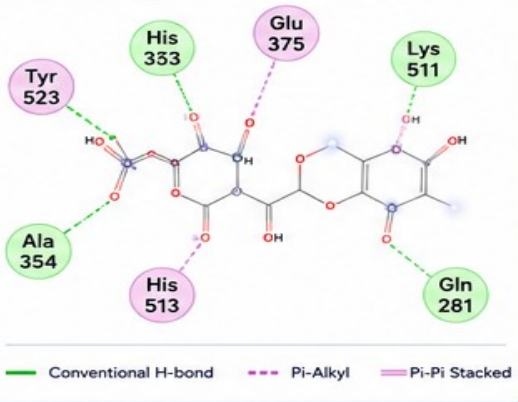
BINDING AFFINITY (kcal/mol) COMPARISON



COMMON KEY INTERACTING RESIDUES

Most of the ligands (phytochemicals and standard drugs) interacted with the following key catalytic residues of ACE active site:

- His353
- Glu375
- Lys511
- Tyr523
- Ala354
- His513
- Gln281



BEST PERFORMER

RUTIN

Binding Affinity: -10.3 kcal/mol

No. of H-bonds: 6

Rutin exhibited the highest binding affinity among all tested compounds, indicating the strongest inhibitory potential against ACE.

Rutin > Isoquercetin > Astragalin > Chlorogenic Acid > Niaziminic Acid > Kaempferol > Lisinopril > Enalapril > Captopril

KEY TAKEAWAYS

- All *Moringa oleifera* phytochemicals showed strong binding affinity with ACE, comparable to standard drugs.
- Rutin (-10.3 kcal/mol) and Isoquercetin (-9.4 kcal/mol) showed better affinity than all standard drugs.
- Multiple hydrogen bonds and key residue interactions contribute to the stability of the ligand–ACE complex.
- These results suggest that *Moringa oleifera* phytochemicals have promising potential as natural ACE inhibitors.

Moringa oleifera phytochemicals, especially Rutin and Isoquercetin, demonstrated superior binding affinity and stable interactions with ACE active site residues, supporting their potential as effective natural ACE inhibitors.

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DOCKING RESULTS – 3D INTERACTION VIEWS (V)

3D binding poses of top *Moringa oleifera* phytochemicals and reference drugs in the ACE active site (PDB ID: 1O86).

ACE Active Site (PDB ID: 1O86)

OVERLAY OF ALL LIGANDS IN ACE ACTIVE SITE

- Rutin
- Isoquercetin
- Astragalin
- Kaempferol
- Lisinopril (Native)
- Captopril
- Enalapril
- Chlorogenic Acid
- Niaziminic Acid

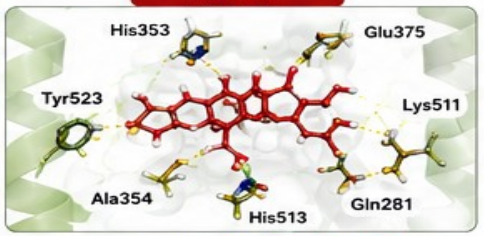


OBSERVATION

- All ligands occupy the ACE catalytic pocket with similar binding orientations.
- Multiple hydrogen bonds and hydrophobic interactions stabilize the ligand–ACE complex.
- Key catalytic residues (His353, Glu375, Lys511, Tyr523, Ala354, His513, Gln281) are consistently involved in interactions.
- Rutin shows the strongest binding affinity (−10.3 kcal/mol) among all tested compounds.

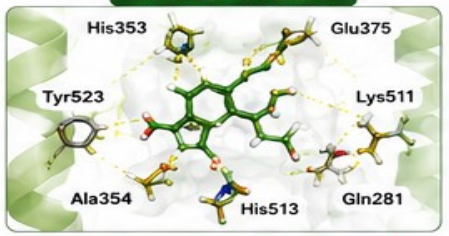
INDIVIDUAL 3D INTERACTION VIEWS

RUTIN



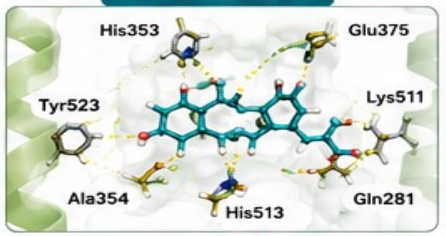
Binding Affinity: −10.3 kcal/mol

ISOQUERCETIN



Binding Affinity: −9.4 kcal/mol

ASTRAGALIN



Binding Affinity: −9.0 kcal/mol

CHLOROGENIC ACID



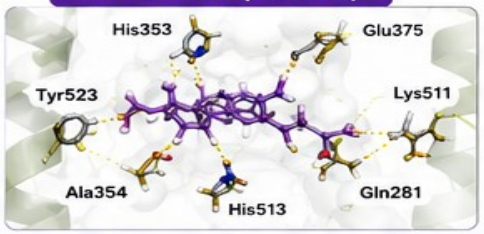
Binding Affinity: −8.8 kcal/mol

KAEMPFEROL



Binding Affinity: −8.2 kcal/mol

LISINOPRIL (NATIVE)



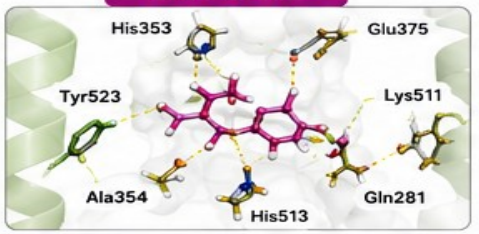
Binding Affinity: −9.6 kcal/mol

ENALAPRIL



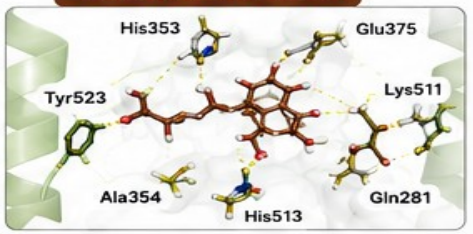
Binding Affinity: −8.6 kcal/mol

CAPTOPRIL



Binding Affinity: −6.7 kcal/mol

CHLOROGENIC ACID



Binding Affinity: −8.0 kcal/mol

CONCLUSION: The 3D interaction analysis confirms that *Moringa oleifera* phytochemicals, especially **Rutin**, bind strongly within the ACE active site through multiple stabilizing interactions, supporting their potential as natural ACE inhibitors.

Hydrogen Bond

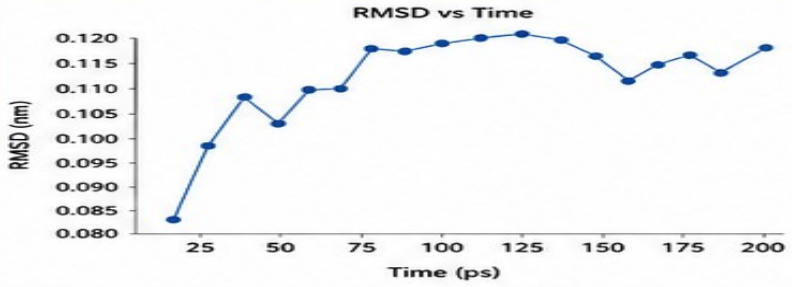
Hydrophobic Interaction

π–π Stacking

MOLECULAR DYNAMICS (MD) SIMULATION ANALYSIS

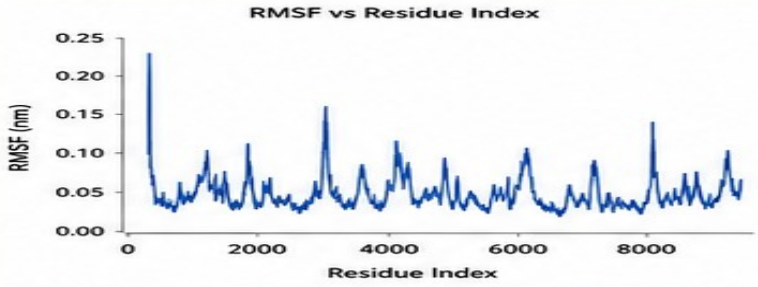
MD simulation was performed on the best docked ACE–ligand complex using GROMACS to evaluate the stability and dynamic behavior of the complex over 200 ps.

1. ROOT MEAN SQUARE DEVIATION (RMSD)



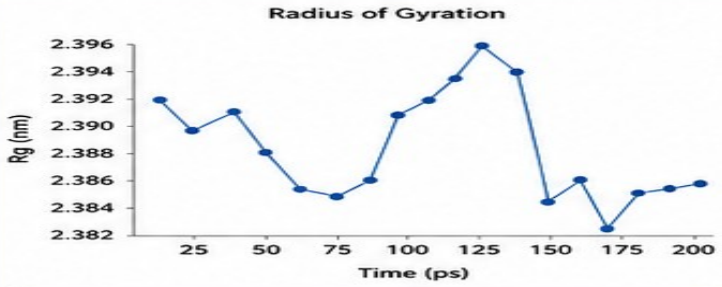
The RMSD plot measures the overall structural deviation of the protein backbone during the 200 ps simulation. RMSD values ranged from **0.082 nm to 0.121 nm**, with a final RMSD recorded as **0.117 nm**.

2. ROOT MEAN SQUARE FLUCTUATION (RMSF)



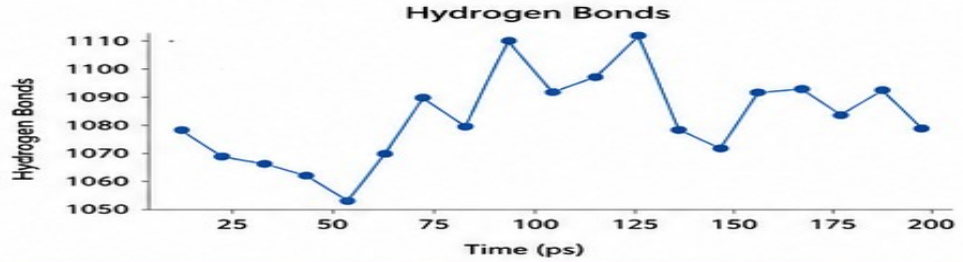
The RMSF analysis evaluates the flexibility of individual residues. Most residues exhibited low fluctuation (**<0.2 nm**), indicating limited atomic mobility. Slightly higher fluctuations were observed in loop regions and terminal residues.

3. RADIUS OF GYRATION (Rg)



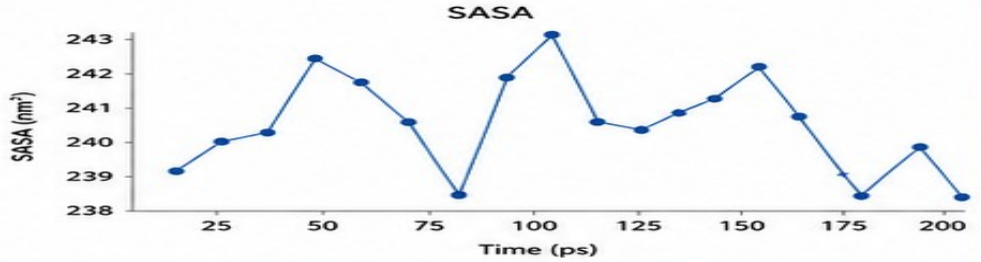
The Radius of Gyration (Rg) reflects the compactness of the protein structure. Rg values fluctuated within **~2.05–2.15 nm**, suggesting a compact and stable structure throughout the simulation.

4. VARIATION IN THE NUMBER OF HYDROGEN BONDS



Hydrogen bonds play a crucial role in stabilizing protein–ligand interactions. During the simulation, the number of hydrogen bonds remained relatively consistent, fluctuating between approximately **1054 and 1116**.

5. SOLVENT ACCESSIBLE SURFACE AREA (SASA)



The SASA analysis estimates the solvent-exposed surface area of the protein. SASA values ranged between approximately **238 and 244 nm²**, with an average around **~200 nm²**.

OVERALL MD ANALYSIS

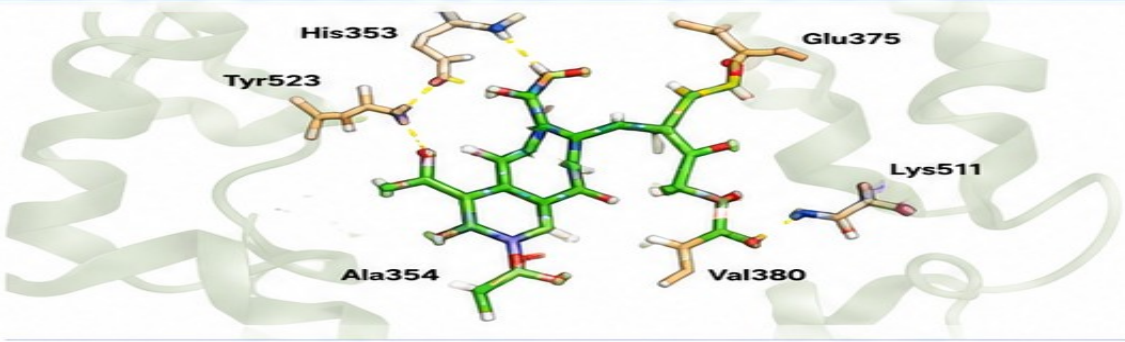
- RMSD values stabilized within **0.082–0.121 nm**, confirming the overall structural stability of the protein–ligand complex.
- RMSF values remained mostly below **0.10 nm**, indicating limited residue flexibility and stable interactions.
- Radius of Gyration (Rg) remained stable around **2.382–2.396 nm**, demonstrating a compact and well-folded structure.
- The number of hydrogen bonds fluctuated between **1054** and **1116**, confirming persistent and stable binding interactions.
- SASA values stayed within **238–244 nm²**, indicating no significant unfolding or structural disruption during simulation.

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MD SIMULATION – INTERACTION AND STABILITY ANALYSIS

Post-MD analysis of the best docked ACE–ligand complex to assess key intermolecular interactions and confirm the stability of the complex throughout the **200 ps** simulation.

1. PROTEIN–LIGAND COMPLEX AFTER 200 ps MD SIMULATION



The ligand remained firmly bound in the ACE active site throughout the simulation, maintaining interactions with key catalytic residues.

2. KEY INTERACTIONS MAINTAINED DURING MD SIMULATION

Residue	Interaction Type	Initial (Docking)	After 200 ps MD	Status
His353	Hydrogen Bond	Present	Maintained	✓
Glu375	Hydrogen Bond	Present	Maintained	✓
Tyr523	Hydrogen Bond / $\pi-\pi$ Interaction	Present	Maintained	✓
Lys511	Hydrogen Bond	Present	Maintained	✓
Ala354	Hydrophobic Interaction	Present	Maintained	✓
Val380	Hydrophobic Interaction	Present	Maintained	✓

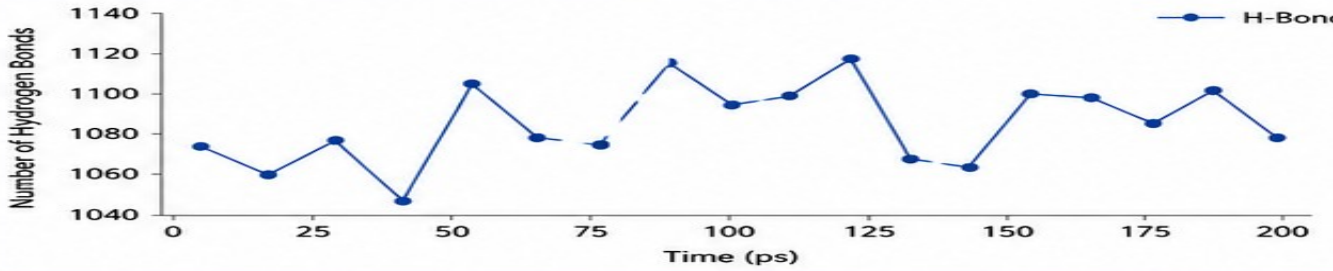
All critical interactions observed in docking were retained throughout the MD simulation, indicating stable binding of the ligand within the ACE active site.

3. MD STABILITY SUMMARY (200 ps SIMULATION)

STABILITY INDICATORS

- RMSD stabilized between **0.082–0.121 nm**, confirming structural stability.
- RMSF values remained mostly below **0.10 nm**, indicating limited residue flexibility.
- Radius of Gyration (Rg) fluctuated within **2.382–2.396 nm**, showing compact structure.
- Hydrogen bonds fluctuated between **1054 and 1116**, confirming persistent interactions.
- SASA values ranged between **238–244 nm²**, indicating no significant unfolding.

INTERACTION PERSISTENCE OVER TIME



The number of hydrogen bonds remained relatively constant throughout the 200 ps simulation, fluctuating between ~**1054** and **1116**.

4. CONCLUSION

- MD simulation over 200 ps confirms that the ACE–ligand complex is structurally stable and well-equilibrated.
- Low RMSD and RMSF values indicate minimal deviation and limited residue flexibility.
- Stable Radius of Gyration and SASA values demonstrate that the protein maintained its compact fold without unfolding.
- Consistent number of hydrogen bonds and retention of key interactions with catalytic residues (His353, Glu375, Tyr523, Lys511) validate the stable binding of the ligand in the ACE active site.
- Overall, MD analysis supports the docking results and confirms the **good inhibitory potential** of the ligand against ACE.

COMPARATIVE COMPUTATIONAL EVALUATION OF *MORINGA OLEIFERA* PHYTOCHEMICALS VS FDA-APPROVED ACE INHIBITORS

THANK YOU

for your time and attention

I would like to express my sincere gratitude to everyone who supported and guided me throughout this 8-week internship. This experience has been truly enriching and instrumental in strengthening my learning and research skills.

SPECIAL THANKS TO

Ms. Mansi
Mentor, BioTechTrek
for her continuous guidance, support, and valuable feedback throughout the project.

BioTechTrek
for providing the opportunity to work on this valuable project and enhancing my practical and computational skills.

• LEARNING • EXPLORING • INNOVATING •

The journey continues...

Aditi Gupta

School of Biotechnology
Gautam Buddha University

Internship Duration
8 Weeks

Organization
BioTechTrek